

ML Midsem Practice — Part 2

All Past Paper Questions + Full Solutions + Mock Exam (Modules 1–5)

SECTION A: MUST-PREPARE QUESTIONS (with Full Solutions)

TYPE 1: Tom Mitchell Definition / Learning Types 2–4 M

Q1 — Identify Learning Type for 4 Scenarios

For each scenario below, identify the type of machine learning and justify:

- (a) A self-driving car learns to navigate roads by receiving rewards for safe driving and penalties for violations.
- (b) A system classifies product reviews as positive, negative, or neutral using labeled examples.
- (c) A bioinformatics tool groups DNA sequences into clusters without any labels.
- (d) An insurance company predicts the dollar amount of future claims from historical data.

Solution:

- (a) **Reinforcement Learning** — The agent interacts with an environment (road), takes actions (steer, brake), and receives feedback (reward/penalty). No labeled dataset is provided; it learns through trial and error.
- (b) **Supervised Learning — Classification** — Labeled training data (review → sentiment class) is used. Output is a discrete category (positive/negative/neutral).
- (c) **Unsupervised Learning — Clustering** — No labels exist. The algorithm discovers natural groupings in DNA sequences based on similarity.
- (d) **Supervised Learning — Regression** — Historical claims (labeled with dollar amounts) are used. Output is a continuous value (dollar amount).

Q2 — Weather Prediction: E, T, P

A machine learning system predicts tomorrow's weather.

- (a) Using Tom Mitchell's definition, identify E, T, P.
- (b) If predicting rainfall amount, is it regression or classification?
- (c) Name two suitable performance measures.

Solution:

- (a) **E (Experience):** Historical weather data (temperature, humidity, wind speed, pressure, past rainfall).
T (Task): Predict tomorrow's weather conditions (sunny/rainy/cloudy or rainfall amount).
P (Performance): Accuracy of predictions compared to actual weather (e.g., % correct, RMSE of rainfall).
- (b) **Regression** — rainfall amount is a continuous numerical value (e.g., 12.5 mm).
- (c) Two performance measures: (i) **RMSE** (Root Mean Squared Error) — penalizes large errors more; (ii) **MAE** (Mean Absolute Error) — average magnitude of errors.

Q3 — Google AlphaGo Versions

Google's Go-playing AI had two versions: (i) the first version learned from thousands of human expert games; (ii) the second version learned entirely by playing against itself. Identify and justify the learning type for each.

Solution:

- (i) **Supervised Learning** — The system learned from labeled data: board positions (input) mapped to expert moves (output). Human games provided the "correct answers."
- (ii) **Reinforcement Learning** — The system played against itself, receiving rewards (win) or penalties (loss). No human-labeled data was used. It discovered strategies through exploration and exploitation.

VARIANT: Netflix recommends movies based on viewing history. What type of learning? Identify E, T, P.

Answer: This is a mix of **Supervised** (predicting ratings from past ratings — collaborative filtering) and **Unsupervised** (clustering similar users/movies).

E: User viewing history, ratings, movie metadata. **T:** Recommend movies a user is likely to enjoy. **P:** Click-through rate, user satisfaction, RMSE on predicted ratings.

TYPE 2: Data Quality Issues 3–5 M**Q1 — Student Dataset**

Identify all data quality issues in this dataset and suggest resolutions:

ID	Name	Age	Grade	Attendance%	City
1	Rina	20	A	92	Mumbai
2	Dev	210	B	78	Delhi
3	Aarav	21	NULL	85	mumbai
4	Shweta	NULL	C	110	Pune
5	Rina	20	A	92	Mumbai
6	Mansi	19	A	88	Delhi

Solution — 6 Issues Found:

- 1. Duplicate Record (Rows 1 & 5):** Rina appears twice with identical data. *Resolution:* Remove the duplicate row.
- 2. Outlier / Invalid Value (Row 2, Age=210):** Age of 210 is impossible. *Resolution:* Verify source; likely 21. Replace with correct value or treat as missing.
- 3. Missing Value (Row 3, Grade=NULL):** Grade is missing. *Resolution:* Impute using mode of grades, or use model-based imputation, or drop if few such rows.
- 4. Missing Value (Row 4, Age=NULL):** Age is missing. *Resolution:* Impute using mean/median age of the dataset.
- 5. Out-of-Range Value (Row 4, Attendance=110%):** Attendance cannot exceed 100%. *Resolution:* Cap at 100 or investigate data entry error.
- 6. Inconsistent Formatting (Row 3, City="mumbai" vs "Mumbai"):** Case inconsistency. *Resolution:* Standardize casing (e.g., title case for all city names).

Q2 — B.Tech Dataset

Identify at least 5 data quality issues:

ID	Name	CGPA	Branch	DOB	Phone
101	Aishwarya	8.5	CSE	12/03/2002	9876543210
102	Bhargav	12.3	ECE	2002-05-14	98765
103	Chandra	NULL	cse	08/11/2001	9123456789
104	Divya	7.2	ME	25/01/2003	9234567890
102	Bhargav	12.3	ECE	2002-05-14	98765
105	Eshan	-1.5	CSE	NULL	9345678901
106	Francis	6.8	IT	30/02/2002	9456789012

Solution — 7 Issues:

1. **Duplicate Record** (Rows 2 & 5): Bhargav appears twice. Remove duplicate.
2. **Out-of-Range CGPA** (Row 2, 12.3): CGPA scale is 0–10. Verify/correct.
3. **Negative CGPA** (Row 6, -1.5): Invalid. Verify/correct.
4. **Missing Values**: CGPA (Row 3), DOB (Row 6). Impute or investigate.
5. **Inconsistent Date Formats**: DD/MM/YYYY vs YYYY-MM-DD. Standardize to one format.
6. **Inconsistent Casing** (Row 3, "cse" vs "CSE"): Standardize to uppercase.
7. **Invalid Date** (Row 7, 30/02/2002): February 30 does not exist. Correct the date.
8. **Invalid Phone** (Row 2, "98765"): Too short. Verify/complete.

VARIANT: Hospital dataset has: (i) missing blood type, (ii) duplicate patient IDs, (iii) age = -5, (iv) DOB in DD/MM/YYYY and MM-DD-YYYY mixed, (v) height recorded as both 170cm and 67in.

Issues: (i) Missing value — impute or flag. (ii) Duplicate — merge or remove. (iii) Invalid/out-of-range — cannot be negative. (iv) Inconsistent format — standardize. (v) Inconsistent units — convert all to one unit (e.g., cm).

TYPE 3: Gradient Descent 6–10 M — MOST IMPORTANT

Q1 [Dec 2025] — Linear Regression + Ridge Regularization

Data: $x = [1, 2, 3]$, $y = [2, 5, 7]$. Initial: $\theta_0 = 0$, $\theta_1 = 1$. Learning rate $\alpha = 0.05$, regularization $\lambda = 2$, $m = 3$.

(a) Write hypothesis and cost function. (b) Perform one iteration of batch gradient descent with L2 regularization. (c) Compare cost before and after.

Solution:**(a) Hypothesis & Cost:**

$$h(x) = \theta_0 + \theta_1 x$$

$$J(\theta) = (1/2m) \sum (h(x^{(i)}) - y^{(i)})^2 + (\lambda/2m) \sum \theta_2^2 \text{ (Ridge: don't regularize } \theta_0)$$

(b) Step-by-step computation:**Step 1 — Predictions:**

$$h(1) = 0 + 1(1) = 1$$

$$h(2) = 0 + 1(2) = 2$$

$$h(3) = 0 + 1(3) = 3$$

Step 2 — Errors ($h(x) - y$):

$$e_1 = 1 - 2 = -1$$

$$e_2 = 2 - 5 = -3$$

$$e_3 = 3 - 7 = -4$$

Step 3 — Gradients:

$$\partial J / \partial \theta_0 = (1/m) \sum e_i = (1/3)(-1 - 3 - 4) = (1/3)(-8) = -2.667$$

$$\begin{aligned} \partial J / \partial \theta_1 &= (1/m) \sum e_i \cdot x_i + (\lambda/m) \theta_1 \\ &= (1/3)[(-1)(1) + (-3)(2) + (-4)(3)] + (2/3)(1) \\ &= (1/3)[-1 - 6 - 12] + 2/3 \\ &= (1/3)(-19) + 2/3 = -19/3 + 2/3 = -17/3 \approx -5.667 \end{aligned}$$

Step 4 — Update:

$$\theta_0 = 0 - 0.05(-2.667) = 0 + 0.133 = 0.133$$

$$\theta_1 = 1 - 0.05(-5.667) = 1 + 0.283 = 1.283$$

(c) Cost comparison:

$$\text{Before: } J = (1/6)[1 + 9 + 16] + (2/6)(1) = 26/6 + 2/6 = 28/6 \approx 4.667$$

$$\text{After: New predictions: } h(1)=0.133+1.283=1.416, h(2)=0.133+2.566=2.699, h(3)=0.133+3.849=3.982$$

$$\text{Errors: } -0.584, -2.301, -3.018$$

$$J = (1/6)(0.341 + 5.295 + 9.108) + (2/6)(1.283)^2 = 14.744/6 + 2(1.646)/6 = 2.457 + 0.549 = 3.006$$

Cost decreased from 4.667 to 3.006. Gradient descent is working.

Q2 [Jan 2026] — Logistic Regression with IPL Data

IPL dataset with 5 teams. Features: Avg_SR (x_1), Match_Wins (x_2), NRR (x_3). Target: Playoff (1/0).
 Team C has missing Avg_SR. After mean imputation, Avg_SR for C = 6.
 After z-score normalization, data is standardized.
 $\theta = [0.1, 0.1, 0.1, 0.2]$, $\alpha = 0.1$. Perform batch gradient descent. Predict probability for Team D.

Solution (Conceptual Steps):

Step 1 — Mean Imputation: Avg_SR values for other teams sum $\div 4 = 6$. Replace Team C's missing value with 6.

Step 2 — Z-score Normalization: For each feature: $x' = (x - \mu) / \sigma$. Compute mean and standard deviation of each column, then standardize.

Step 3 — Sigmoid & Predictions:

$$z = \theta_0 + \theta_1 x_1' + \theta_2 x_2' + \theta_3 x_3'$$

$$\sigma(z) = 1 / (1 + e^{-z})$$

Step 4 — Gradients (Logistic Regression):

$$\partial J / \partial \theta_j = (1/m) \sum (\sigma(z^{(i)}) - y^{(i)}) \cdot x_j^{(i)}$$

Step 5 — Update: $\theta_j = \theta_j - \alpha \cdot \partial J / \partial \theta_j$

Step 6 — Cross-Entropy Loss:

$$J = -(1/m) \sum [y \log(\sigma(z)) + (1-y) \log(1-\sigma(z))]$$

Step 7 — Prediction for Team D:

After plugging in Team D's normalized features and computing the sigmoid, the predicted probability is approximately **77.5%**. Since this exceeds the 0.5 threshold, Team D is predicted to make the playoffs.

Q3 [Revision] — Compute $\partial J / \partial \theta_1$ Only

Data: $x = [2, 4, 7, 8, 10]$, $y = [1, 2, 2.5, 3.5, 5.5]$. $\theta_0 = 0$, $\theta_1 = 0.5$. Compute $\partial J / \partial \theta_1$.

Solution:**Step 1 — Predictions:** $h(x) = 0 + 0.5x$

$$h(2) = 1, h(4) = 2, h(7) = 3.5, h(8) = 4, h(10) = 5$$

Step 2 — Errors (h - y):

$$1-1 = 0, \quad 2-2 = 0, \quad 3.5-2.5 = 1, \quad 4-3.5 = 0.5, \quad 5-5.5 = -0.5$$

Step 3 — Error $\times$ x:

$$0 \times 2 = 0, \quad 0 \times 4 = 0, \quad 1 \times 7 = 7, \quad 0.5 \times 8 = 4, \quad (-0.5) \times 10 = -5$$

Step 4 — Sum and divide:

$$\text{Sum} = 0 + 0 + 7 + 4 + (-5) = 6$$

$$\partial J / \partial \theta_1 = 6/5 = \mathbf{1.2}$$

VARIANT: $x = [1, 3, 5], y = [2, 6, 8], \theta_0 = 1, \theta_1 = 1, \alpha = 0.1$, no regularization. Perform 1 iteration.**Solution:**

$$\text{Predictions: } h(1)=2, h(3)=4, h(5)=6$$

$$\text{Errors: } 2-2=0, 4-6=-2, 6-8=-2$$

$$\partial J / \partial \theta_0 = (1/3)(0-2-2) = -4/3 \approx -1.333$$

$$\partial J / \partial \theta_1 = (1/3)(0 \times 1 + (-2) \times 3 + (-2) \times 5) = (1/3)(-16) = -16/3 \approx -5.333$$

$$\theta_0 = 1 - 0.1(-1.333) = 1.133$$

$$\theta_1 = 1 - 0.1(-5.333) = 1.533$$

TYPE 4: Confusion Matrix 5-6 M**Q1 [Dec 2025] — Medical Diagnosis**

1000 patients tested. 150 actually have the disease. Model: TP = 120, FP = 85. Build the confusion matrix and compute all metrics. If a missed diagnosis costs \$10,000 and a false alarm costs \$500, compute total cost.

Solution:**Derive missing values:**

Actual Positive = 150 $\Rightarrow$ FN = 150 - 120 = 30

Actual Negative = 1000 - 150 = 850 $\Rightarrow$ TN = 850 - 85 = 765

	Predicted +	Predicted -	Total
Actual +	TP = 120	FN = 30	150
Actual -	FP = 85	TN = 765	850
Total	205	795	1000

Metrics:

Accuracy = (120 + 765) / 1000 = 885/1000 = **88.5%**

Precision = 120 / (120 + 85) = 120/205 = **58.5%**

Recall (Sensitivity) = 120 / (120 + 30) = 120/150 = **80.0%**

Specificity = 765 / (765 + 85) = 765/850 = **90.0%**

F1-Score = $2 \times (0.585 \times 0.80) / (0.585 + 0.80) = 2(0.468) / 1.385 = \mathbf{0.676}$

Cost Analysis:

Cost = FN $\times$ \$10,000 + FP $\times$ \$500 = 30 $\times$ 10,000 + 85 $\times$ 500 = \$300,000 + \$42,500 = **\$342,500**

Since missed diagnoses are 20 $\times$ more costly, the model should prioritize higher recall (lower threshold) even at the expense of more false positives.

Q2 [Jan 2026] — Two Models with Posterior Probabilities

10 instances. Actual labels and two models' posterior probabilities given. Threshold = 0.5. Build confusion matrix for each model, compute Precision, Recall, F1.

Solution approach:

Step 1: For each instance, if $P(\text{positive}) \geq 0.5 \Rightarrow$ predict positive, else negative.

Step 2: Compare predictions to actual labels to fill confusion matrix.

Step 3: Compute metrics for each model.

Step 4: The model with higher F1-score is generally better. If domain requires high recall (medical), pick the model with higher recall even if precision drops.

Key insight: Threshold of 0.5 is default. Adjusting threshold trades off precision vs recall. Lower threshold $\Rightarrow$ more positives predicted $\Rightarrow$ higher recall, lower precision.

Q3 [2024 Regular] — Metrics for NEGATIVE Class

TP = 350, FN = 150, FP = 200, TN = 800. Compute Precision, Recall, F1 for the *negative* class. When is precision > recall? When is recall > precision?

Solution:

For the **negative class**, flip the perspective: what was "negative" becomes our "target."

$TP_{\text{neg}} = TN = 800$, $FP_{\text{neg}} = FN = 150$, $FN_{\text{neg}} = FP = 200$, $TN_{\text{neg}} = TP = 350$

$\text{Precision}_{\text{neg}} = 800 / (800 + 150) = 800/950 = \mathbf{84.2\%}$

$\text{Recall}_{\text{neg}} = 800 / (800 + 200) = 800/1000 = \mathbf{80.0\%}$

$F1_{\text{neg}} = 2(0.842 \times 0.80) / (0.842 + 0.80) = 1.347 / 1.642 = \mathbf{0.820}$

Precision > Recall when the model is conservative — it predicts positive only when very confident, so few false positives but may miss some actual positives (FN is high).

Recall > Precision when the model is aggressive — it predicts positive liberally, catching most actual positives but also flagging many negatives (FP is high). Common in medical screening.

Q4 [2024 Makeup] — 3 Models for Medical Diagnosis

Three models for disease detection. TP, FP, FN, TN given for each. Which model should be chosen for medical diagnosis?

Solution approach:

Compute Precision, Recall, F1 for each model. For **medical diagnosis**, recall is critical because a missed disease (FN) can be fatal. Choose the model with the **highest recall**, provided precision is acceptable. If two models have similar recall, prefer the one with higher F1. Never optimize for accuracy alone in imbalanced medical datasets.

VARIANT: E-commerce fraud: 5000 transactions, 200 fraudulent. Model: TP=150, FP=300.

$FN = 200 - 150 = 50$, $TN = 4800 - 300 = 4500$.

Precision = $150/450 = 33.3\%$, Recall = $150/200 = 75.0\%$, F1 = 0.461

Accuracy = $4650/5000 = 93\%$ (misleading due to imbalance!)

Cost: Missed fraud = $50 \times \$10,000 = \$500,000$. Investigation = $300 \times \$100 = \$30,000$. Total = \$530,000.

Recommendation: Lower threshold to catch more fraud (increase recall). The \$100 investigation cost is negligible compared to \$10K fraud loss.

TYPE 5: Bias-Variance 3–6 M**Q1 [Dec 2025] — Two Polynomial Models**

Model A (degree 2): Train error=15, Test error=18. Model B (degree 15): Train error=0.5, Test error=45.

Learning curves show B's validation error increasing. Manager says: "Use higher order polynomial."

Evaluate.

Solution:

Model A (degree 2): Train $\approx$ Test $\Rightarrow$ **High Bias (Underfitting)**. The model is too simple to capture the data pattern. Both errors are moderately high.

Model B (degree 15): Train ≈ 0 but Test is very high $\Rightarrow$ **High Variance (Overfitting)**. The model memorizes training data but fails on unseen data. Large gap between train and test error.

Manager's suggestion is WRONG. Higher-order polynomials will make overfitting worse. Instead:

- Use regularization (L1/L2) to penalize large coefficients
- Get more training data
- Use cross-validation to select the optimal polynomial degree
- A moderate degree (e.g., 4–6) would likely be best

Q2 [2024 Makeup] — 4 Models with RMSE/R²

Model	Train RMSE	Test RMSE	R ² (Test)
A	2.1	2.3	0.92
B	0.1	15.4	0.10
C	8.5	8.7	0.35
D	3.0	3.2	0.85

Diagnose each model. Why does B diverge? What is R²?

Solution:

Model A: Low train & test RMSE, high R^2 . **Good fit.** Best model overall.

Model B: Near-zero train error, huge test error, low R^2 . **Severe overfitting (High Variance).**

Model C: Both errors high, low R^2 . **Underfitting (High Bias).** Too simple.

Model D: Moderate errors, decent R^2 . **Reasonable fit** but slightly worse than A.

Why B diverges: The model is too complex (e.g., high-degree polynomial or too many features without regularization). It memorizes noise in training data, leading to wild predictions on test data.

R^2 (Coefficient of Determination): $R^2 = 1 - (SS_{\text{res}} / SS_{\text{tot}})$. Measures the proportion of variance explained by the model. $R^2 = 1$ means perfect fit, $R^2 = 0$ means no better than predicting the mean. R^2 can be negative if the model is worse than the mean baseline.

Q3 [Jan 2026] — "Doubled Data But Validation Still High"

A student says: "I doubled my training data, but validation error is still high." Critique this statement.

Solution:

If validation error is high *and* training error is also high $\Rightarrow$ **High Bias problem.** Adding more data does NOT help with high bias. The model is too simple to learn the pattern regardless of data size.

Fixes for High Bias:

- Increase model complexity (more features, higher degree, deeper network)
- Use a more powerful model (e.g., switch from linear to polynomial or ensemble)
- Add meaningful features or feature engineering
- Decrease regularization (if too strong)

Adding more data only helps when the problem is **high variance** (train error low, validation error high).

VARIANT: Model X: train acc = 0.99, test acc = 0.52. Model Y: train = 0.72, test = 0.70.

Model X: High variance (overfitting). Huge gap. Fix: regularization, more data, simpler model, dropout.

Model Y: High bias (underfitting). Both accuracies are low. Fix: more complex model, more features, less regularization.

Deploy: Neither as-is. But Model Y is more stable/predictable. Fix X first (easier to tame an overfit model), or increase Y's complexity. In production, stability matters — Y is safer to start with.

TYPE 6: Regularization 2–4 M

Q1 — L1 vs L2

Compare L1 and L2 regularization. Which is better for a dataset with many irrelevant features?

Solution:

Aspect	L1 (Lasso)	L2 (Ridge)
Penalty term	$\lambda \sum \theta_j $	$\lambda \sum \theta_j^2$
Effect on weights	Drives some weights to exactly 0	Shrinks weights toward 0 but not to 0
Feature selection	YES — built-in	NO — keeps all features
Sparse solution	Yes	No
When multicollinearity	Picks one feature, drops rest	Distributes weight among correlated features

For many irrelevant features: Use **L1 (Lasso)** because it performs automatic feature selection by setting irrelevant feature weights to exactly zero, producing a simpler, more interpretable model.

Q2 [2024 Makeup] — Overfitting Logistic Regression

A logistic regression model has very large coefficients and overfits the training data. Diagnose and suggest fixes.

Solution:

Diagnosis: Large coefficients indicate the model is fitting noise in the training data. The decision boundary becomes overly complex. This is **high variance / overfitting**.

Fixes:

1. **Add L2 Regularization (Ridge):** Penalize large coefficients: $J = \text{original_cost} + (\lambda/2m)\sum\theta^2$. Forces weights to be small.
2. **Add L1 Regularization (Lasso):** If many features are irrelevant, L1 will zero them out.
3. **Reduce number of features** (feature selection/PCA).
4. **Increase training data** to reduce variance.
5. **Increase λ** (regularization strength) — tune via cross-validation.

TYPE 7: Decision Tree / Entropy 5–8 M**Q1 [Dec 2025] — Information Gain Calculation**

14 samples: 9 Approved (R), 5 Denied (D).
Feature "Income": High (6R, 2D), Low (3R, 3D).
Compute Information Gain for splitting on Income.

Solution:**Step 1 — Parent Entropy:**

$$\begin{aligned}
 H(S) &= -(9/14)\log_2(9/14) - (5/14)\log_2(5/14) \\
 &= -(9/14)(-0.637) - (5/14)(-1.485) \\
 &= 0.410 + 0.530 \\
 &= \mathbf{0.940}
 \end{aligned}$$

Step 2 — Child Entropies:

High Income (8 samples: 6R, 2D):

$$\begin{aligned}
 H(\text{High}) &= -(6/8)\log_2(6/8) - (2/8)\log_2(2/8) \\
 &= -(0.75)(-0.415) - (0.25)(-2.0) \\
 &= 0.311 + 0.500 = \mathbf{0.811}
 \end{aligned}$$

Low Income (6 samples: 3R, 3D):

$$\begin{aligned}
 H(\text{Low}) &= -(3/6)\log_2(3/6) - (3/6)\log_2(3/6) \\
 &= -(0.5)(-1) - (0.5)(-1) \\
 &= 0.5 + 0.5 = \mathbf{1.000} \text{ (maximum entropy — pure uncertainty)}
 \end{aligned}$$

Step 3 — Weighted Average Entropy:

$$\begin{aligned}
 H(S|\text{Income}) &= (8/14) \times 0.811 + (6/14) \times 1.000 \\
 &= 0.463 + 0.429 = \mathbf{0.892}
 \end{aligned}$$

Step 4 — Information Gain:

$$IG(S, \text{Income}) = H(S) - H(S|\text{Income}) = 0.940 - 0.892 = \mathbf{0.048}$$

This is a relatively small gain. We would compare with other features and pick the one with the highest IG as the root node.

Q2 [Jan 2026] — Customer Data Decision Tree

8 customers. Features: AgeGroup (Young/Old), Income (High/Low), Occupation (Salaried/Self-Employed).
Target: Buy (Yes/No).

(a) Compute entropy of target. (b) Compute IG for AgeGroup. (c) If Young branch has mixed labels, what is the best next split? (d) If a leaf has 3 Yes, 1 No, what class is assigned?

Solution (Illustrative with 5 Yes, 3 No):**(a) Parent Entropy:**

$$\begin{aligned}
 H(S) &= -(5/8)\log_2(5/8) - (3/8)\log_2(3/8) \\
 &= -(0.625)(-0.678) - (0.375)(-1.415) \\
 &= 0.424 + 0.530 = \mathbf{0.954}
 \end{aligned}$$

(b) IG for AgeGroup (assume Young:4 samples, Old:4 samples):

Compute $H(\text{Young})$ and $H(\text{Old})$, then weighted average. $IG = H(S) - \text{weighted average}$.

(c) Best next split: Compute IG for Income and Occupation within the Young subset. Pick the feature with the highest IG.

(d) Majority Voting: 3 Yes vs 1 No $\Rightarrow$ Assign class **"Yes"**. In decision trees, leaf nodes use majority class when the branch is not pure.

Q3 [2024 Regular] — 15 Students, 3 Classes

15 students: 5 in class A, 5 in class B, 5 in class C. Three features available. Find the root node.

Solution approach:

Parent Entropy (3-class, uniform):

$$H(S) = -3 \times (5/15)\log_2(5/15) = -3 \times (1/3)\log_2(1/3) = -3 \times (1/3)(-1.585) = \mathbf{1.585 \text{ bits}}$$

This is $\log_2(3)$ = maximum entropy for 3 classes (uniform distribution).

Compute IG for each of the 3 features. The feature with the **highest IG** becomes the root node. The root node is the feature that best separates the classes (reduces uncertainty the most).

Q4 [2024 Makeup] — Information Theory Concepts

- (a) How many bits needed to encode 4 equally likely classes?
- (b) What is entropy of a uniform distribution over 4 classes?
- (c) What happens to entropy when distribution becomes non-uniform?
- (d) Compare information content of an event with $P=0.01$ vs $P=0.99$.

Solution:

(a) Bits = $\log_2(4) = \mathbf{2 \text{ bits}}$. With 2 bits we can represent 4 states: 00, 01, 10, 11.

(b) $H = -4 \times (1/4) \log_2(1/4) = -4 \times (1/4)(-2) = \mathbf{2.0 \text{ bits}}$. Maximum entropy equals $\log_2(\text{number of classes})$ when distribution is uniform.

(c) Entropy **decreases**. Non-uniform means some outcomes are more likely, reducing uncertainty. E.g., [0.7, 0.1, 0.1, 0.1] has $H \approx 1.36 < 2.0$. Entropy is maximized only for uniform distributions.

(d) Information content $I(x) = -\log_2(P(x))$.

$I(0.01) = -\log_2(0.01) = \mathbf{6.64 \text{ bits}}$ (rare event $\Rightarrow$ very informative)

$I(0.99) = -\log_2(0.99) = \mathbf{0.014 \text{ bits}}$ (common event $\Rightarrow$ almost no information)

Rare events carry much more information than common ones.

VARIANT: 20 emails: 12 spam (S), 8 ham (H). Feature "contains_link": Yes (10S, 2H), No (2S, 6H). Compute IG.

Parent Entropy: $H(S) = -(12/20) \log_2(12/20) - (8/20) \log_2(8/20) = -(0.6)(-0.737) - (0.4)(-1.322) = 0.442 + 0.529 = \mathbf{0.971}$

Yes branch (12 samples: 10S, 2H):

$H(\text{Yes}) = -(10/12) \log_2(10/12) - (2/12) \log_2(2/12) = -(0.833)(-0.263) - (0.167)(-2.585) = 0.219 + 0.431 = \mathbf{0.650}$

No branch (8 samples: 2S, 6H):

$H(\text{No}) = -(2/8) \log_2(2/8) - (6/8) \log_2(6/8) = -(0.25)(-2) - (0.75)(-0.415) = 0.500 + 0.311 = \mathbf{0.811}$

Weighted: $H(S|\text{link}) = (12/20)(0.650) + (8/20)(0.811) = 0.390 + 0.324 = \mathbf{0.714}$

$IG = 0.971 - 0.714 = \mathbf{0.257}$ — a good split.

TYPE 8: Logistic Regression Concepts 2–5 M**Q1 — Large Coefficient Interpretation**

In a college admission model, θ_2 (for CGPA) is very large. Does a high CGPA guarantee selection?

Answer: NO.

A large θ_2 means CGPA has a **strong influence** on the prediction, but it does not guarantee selection because:

1. The sigmoid function outputs a **probability** (0 to 1), never a certainty.
2. Other features (θ_1 , θ_3 , etc.) and the bias (θ_0) also contribute. Negative values of other weighted features can offset CGPA's contribution.
3. The total $z = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots$ must be evaluated. Even with large θ_2 -CGPA, if other terms are strongly negative, z could be negative $\Rightarrow \sigma(z) < 0.5 \Rightarrow$ rejected.

Q2 — Coefficient Interpretation & Unit Change

- (a) What does a coefficient of 0.8 for "years of experience" mean?
- (b) If experience is measured in months instead of years, what happens to the coefficient?
- (c) What does a negative gradient for a weight mean?

(a) For every 1-unit increase in years of experience, the log-odds of the positive class increases by 0.8. Equivalently, the odds are multiplied by $e^{0.8} \approx 2.23$.

(b) If measured in months (1 year = 12 months), the coefficient becomes $0.8/12 \approx \mathbf{0.067}$. The model's predictions remain the same; only the coefficient scale changes. This is why feature scaling matters for interpretation and convergence.

(c) A negative gradient $\partial J / \partial \theta_j < 0$ means increasing θ_j would **decrease** the cost. Therefore, gradient descent (which subtracts $\alpha \cdot$ gradient) will **increase** θ_j : $\theta_j = \theta_j - \alpha(\text{negative}) = \theta_j + \text{positive value}$.

Q3 — Decision Boundary Shape

Weight vector $w = \langle -36, 0, 0, 4, 9 \rangle$ for features $[1, x_1, x_2, x_1^2, x_2^2]$. What is the shape of the decision boundary?

Solution:

Decision boundary: $w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2 = 0$

$$-36 + 0 \cdot x_1 + 0 \cdot x_2 + 4x_1^2 + 9x_2^2 = 0$$

$$4x_1^2 + 9x_2^2 = 36$$

$$x_1^2/9 + x_2^2/4 = 1$$

This is an **ellipse** centered at the origin with semi-major axis 3 along x_1 and semi-minor axis 2 along x_2 .

Points inside the ellipse belong to one class and outside to the other. Polynomial features enable non-linear decision boundaries in logistic regression.

TYPE 9: Feature Scaling True/False 6 M

State True or False with justification:

- (a) Feature scaling is always necessary for gradient descent.
- (b) Decision trees require feature scaling.
- (c) Min-max scaling is sensitive to outliers.
- (d) Z-score normalization always produces values between 0 and 1.

(a) TRUE. Without scaling, features with larger ranges dominate the gradient, causing slow or oscillating convergence. Gradient descent converges much faster when features are on similar scales. (Technically "strongly recommended" rather than "always necessary," but for exam purposes: True.)

(b) FALSE. Decision trees split on thresholds for individual features. They compare values within a feature, not across features. Scaling does not change the split points or the tree structure. Trees are scale-invariant.

(c) TRUE. Min-max scaling: $x' = (x - x_{\min}) / (x_{\max} - x_{\min})$. A single extreme outlier can drastically change $x_{\min}$ or $x_{\max}$, compressing all other values into a narrow range.

(d) FALSE. Z-score normalization: $x' = (x - \mu) / \sigma$. It produces values with mean 0 and std 1, but values can range from $-\infty$ to $+\infty$. A value 3 standard deviations above mean gives $x' = 3$, which is not in $[0,1]$.

TYPE 10: Model Robustness 6 M

Three models tested on two datasets — one with noise, one with outliers. For each model and dataset, state whether it is Affected or Less Affected, with justification.

Models: Linear Regression, Decision Tree, Random Forest.

Model	Noise	Outliers
Linear Regression	Affected	Affected
Decision Tree	Affected	Less Affected
Random Forest	Less Affected	Less Affected

Justifications:

Linear Regression + Noise: Affected. LR minimizes squared errors, so noisy data shifts the line. Every point influences the fit.

Linear Regression + Outliers: Affected. Squared error amplifies outlier influence (error² grows quadratically). A single outlier can dramatically shift the regression line.

Decision Tree + Noise: Affected. Decision trees tend to overfit noise by creating overly specific splits to accommodate noisy data points.

Decision Tree + Outliers: Less Affected. Trees split on thresholds; outliers typically end up in their own leaf without affecting splits elsewhere.

Random Forest + Noise: Less Affected. Bagging (averaging many trees) smooths out noise. Each tree sees a different bootstrap sample, so noise averages out.

Random Forest + Outliers: Less Affected. Similar reasoning — outliers affect only a subset of trees, and averaging reduces their impact.

TYPE 11: Normal Equation Dimensions 2 M

$m = 28$ (samples), $n = 4$ (features). What are the dimensions of X , y , and θ in the normal equation $\theta = (X^T X)^{-1} X^T y$?

Solution:

X (with bias column $x_0=1$): 28×5 ($m \times (n+1)$)

y : 28×1 ($m \times 1$)

θ : 5×1 ($(n+1) \times 1$)

Intermediate dimensions:

X^T : 5×28

$X^T X$: 5×5 (square matrix, invertible if features are linearly independent)

$(X^T X)^{-1}$: 5×5

$(X^T X)^{-1} X^T$: 5×28

$(X^T X)^{-1} X^T y$: $5 \times 1 = \theta$ ✓

SECTION B: MOCK EXAM — TRY YOURSELF FIRST**ML MIDSEM MOCK PAPER**

Total Marks: 30 | Duration: 2 Hours | Modules 1–5

Answer all questions. Show all working for full marks.

Q1 [5 Marks] — Tom Mitchell & Learning Concepts

A hospital wants to build an ML system to predict whether a discharged patient will be readmitted within 30 days.

(a) Using Tom Mitchell's definition, identify E, T, P. [2M]

(b) Is this classification or regression? What if the hospital wants to predict the number of days until readmission? [1M]

(c) If the model overfits, what are the implications? Suggest two remedies (one must be regularization or early stopping). [2M]

Q2 [6 Marks] — Data Quality Issues

Identify all data quality issues in this hospital patient dataset and suggest resolutions:

PID	Name	Age	BloodType	Admission Date	BP
P001	Arjun	45	O+	15/03/2025	120/80
P002	Bhavna	32	NULL	2025-03-18	130/85
P003	Chirag	-5	A+	20/03/2025	125/82
P004	Diya	28	B+	22/03/2025	normal
P001	Arjun	45	O+	15/03/2025	120/80
P005	Esha	200	AB-	25/03/2025	140/90

Q3 [8 Marks] — Linear Regression Gradient Descent with Regularization

Data: $x = [2, 4, 6]$, $y = [3, 7, 11]$. Initial: $\theta_0 = 0.5$, $\theta_1 = 0.5$. Learning rate $\alpha = 0.1$, regularization $\lambda = 1$, $m = 3$.

- Write the hypothesis function and Ridge-regularized cost function. [2M]
- Perform one iteration of batch gradient descent. Show all intermediate steps. [4M]
- Compute cost before and after the update. Is gradient descent working? [2M]

Q4 [5 Marks] — Confusion Matrix & Cost Analysis

An email spam filter processes 2000 emails: 400 are spam, 1600 are legitimate.

Model results: TP = 350, FP = 100, FN = 50, TN = 1500.

- Draw the confusion matrix. [1M]
- Compute Precision, Recall, F1-Score. [2M]
- Marking a good email as spam costs \$50 (user misses important email). Missing a spam email costs \$5. Compute total misclassification cost. Based on this, should the threshold be raised or lowered? [2M]

Q5 [3 Marks] — Bias-Variance Tradeoff

Model A: Train RMSE = 500, Test RMSE = 520.

Model B: Train RMSE = 50, Test RMSE = 900.

- (a) Diagnose each model. [1M]
- (b) Suggest one specific fix for each. [1M]
- (c) If you had to deploy one model immediately, which would you choose and why? [1M]

Q6 [3 Marks] — Decision Tree / Information Gain

A fruit basket has 10 fruits: 6 Apples (A), 4 Oranges (O).

Feature "Color": Red $\rightarrow$ 5A, 1O; Green $\rightarrow$ 1A, 3O.

- (a) Compute entropy of the parent node. [1M]
- (b) Compute entropy of each child node and the weighted average. [1M]
- (c) Compute Information Gain. Should we split on "Color"? [1M]

ANSWERS BELOW — TRY THE MOCK EXAM FIRST!

MOCK EXAM — FULL SOLUTIONS

Q1 Solution — Tom Mitchell & Learning Concepts [5M]

(a) Tom Mitchell's Definition:

"A computer program learns from experience E with respect to task T and performance measure P if its performance on T, as measured by P, improves with experience E."

E (Experience): Historical patient records — demographics, diagnoses, medications, length of stay, lab results, and whether each patient was readmitted within 30 days.

T (Task): Predict whether a newly discharged patient will be readmitted within 30 days (binary: Yes/No).

P (Performance): Accuracy, F1-score, or AUC-ROC on a held-out test set of patient records.

(b) Classification vs Regression:

Predicting readmission (Yes/No) $\Rightarrow$ **Binary Classification** (discrete output).

Predicting number of days until readmission $\Rightarrow$ **Regression** (continuous numerical output).

(c) Overfitting Implications & Remedies:

Implications: The model memorizes training data patterns (including noise) and fails to generalize. It may achieve 99% accuracy on training data but perform poorly on new patients, leading to unreliable readmission predictions that could misallocate hospital resources.

Remedy 1 — Regularization: Add L2 (Ridge) penalty to the cost function: $J + (\lambda/2m)\Sigma\theta^2$. This constrains model complexity by penalizing large coefficients, preventing the model from fitting noise.

Remedy 2 — Early Stopping: Monitor validation loss during training. Stop training when validation loss begins to increase (even if training loss is still decreasing). This prevents the model from over-learning the training data.

Q2 Solution — Data Quality Issues [6M]

6 Issues Identified:

1. Duplicate Record (Rows 1 & 5): PID P001 (Arjun) appears twice with identical data.

Resolution: Remove the duplicate row. Verify in the source system whether it was a data entry error.

2. Missing Value (Row 2, BloodType = NULL): Bhavna's blood type is missing.

Resolution: Cannot impute blood type statistically (it's categorical with no correlation to other features).

Either obtain from medical records or flag as "Unknown."

3. Invalid Value — Negative Age (Row 3, Age = -5): Age cannot be negative.

Resolution: Verify correct age from records. If unavailable, treat as missing and impute with median age.

4. Inconsistent Data Type (Row 4, BP = "normal"): All other BP values are numerical (e.g., 120/80) but this row has a text value.

Resolution: Standardize BP format. Either get the actual reading or encode as missing. "Normal" is subjective and not a valid measurement.

5. Inconsistent Date Format (Row 2 vs others): Row 2 uses YYYY-MM-DD while others use DD/MM/YYYY.

Resolution: Standardize all dates to a single format (e.g., YYYY-MM-DD, ISO 8601).

6. Outlier / Impossible Value (Row 6, Age = 200): No human lives to 200 years.

Resolution: Verify; likely a data entry error (perhaps 20). Correct or treat as missing.

Q3 Solution — Linear Regression Gradient Descent [8M]**(a) Hypothesis & Cost Function:**

$$h(x) = \theta_0 + \theta_1 x$$

$$J(\theta) = (1/2m) \sum (h(x^{(i)}) - y^{(i)})^2 + (\lambda/2m) \theta_1^2$$

Note: θ_0 (bias) is NOT regularized.

(b) One Iteration of Gradient Descent:**Step 1 — Predictions:**

$$h(2) = 0.5 + 0.5(2) = 0.5 + 1.0 = \mathbf{1.5}$$

$$h(4) = 0.5 + 0.5(4) = 0.5 + 2.0 = \mathbf{2.5}$$

$$h(6) = 0.5 + 0.5(6) = 0.5 + 3.0 = \mathbf{3.5}$$

Step 2 — Errors ($h(x) - y$):

$$e_1 = 1.5 - 3 = -1.5$$

$$e_2 = 2.5 - 7 = -4.5$$

$$e_3 = 3.5 - 11 = -7.5$$

Step 3 — Gradients:

$$\partial J / \partial \theta_0 = (1/m) \sum e_i = (1/3)(-1.5 - 4.5 - 7.5) = (1/3)(-13.5) = -4.5$$

$$\begin{aligned} \partial J / \partial \theta_1 &= (1/m) \sum e_i \cdot x_i + (\lambda/m) \theta_1 \\ &= (1/3)[(-1.5)(2) + (-4.5)(4) + (-7.5)(6)] + (1/3)(0.5) \\ &= (1/3)[-3 - 18 - 45] + 0.167 \\ &= (1/3)(-66) + 0.167 \\ &= -22 + 0.167 = -21.833 \end{aligned}$$

Step 4 — Weight Updates:

$$\theta_0 = 0.5 - 0.1(-4.5) = 0.5 + 0.45 = 0.95$$

$$\theta_1 = 0.5 - 0.1(-21.833) = 0.5 + 2.183 = 2.683$$

(c) Cost Before and After:

Before:

$$\begin{aligned} J &= (1/6)[(-1.5)^2 + (-4.5)^2 + (-7.5)^2] + (1/6)(0.5)^2 \\ &= (1/6)[2.25 + 20.25 + 56.25] + (1/6)(0.25) \\ &= (1/6)(78.75) + 0.042 = 13.125 + 0.042 = 13.167 \end{aligned}$$

After (new $\theta_0 = 0.95$, $\theta_1 = 2.683$):

$$h(2) = 0.95 + 2.683(2) = 6.316$$

$$h(4) = 0.95 + 2.683(4) = 11.682$$

$$h(6) = 0.95 + 2.683(6) = 17.048$$

$$\text{Errors: } 6.316 - 3 = 3.316, \quad 11.682 - 7 = 4.682, \quad 17.048 - 11 = 6.048$$

$$\begin{aligned} J &= (1/6)(10.996 + 21.921 + 36.578) + (1/6)(2.683)^2 \\ &= (1/6)(69.495) + (1/6)(7.198) = 11.583 + 1.200 = 12.783 \end{aligned}$$

Cost decreased from 13.167 to 12.783. **Gradient descent is working**, though the decrease is modest because the learning rate caused a slight overshoot on θ_1 . A smaller α would give smoother convergence.

Q4 Solution — Confusion Matrix [5M]

(a) Confusion Matrix:

	Predicted Spam	Predicted Legit	Total
Actual Spam	TP = 350	FN = 50	400
Actual Legit	FP = 100	TN = 1500	1600
Total	450	1550	2000

(b) Metrics:

Precision = $TP / (TP + FP) = 350 / (350 + 100) = 350/450 = 77.8\%$

Recall = $TP / (TP + FN) = 350 / (350 + 50) = 350/400 = 87.5\%$

F1-Score = $2 \times (0.778 \times 0.875) / (0.778 + 0.875) = 2(0.681) / 1.653 = 1.361 / 1.653 = 0.823$

(c) Cost Analysis:

FP cost: Legit email marked as spam = $100 \times \$50 = \$5,000$

FN cost: Spam email missed = $50 \times \$5 = \250

Total cost = $\$5,000 + \$250 = \$5,250$

Recommendation: RAISE the threshold.

The dominant cost is false positives (\$50 each) vs false negatives (\$5 each) — a 10:1 ratio. FP cost is 20× the FN cost (\$5,000 vs \$250). By raising the threshold (e.g., from 0.5 to 0.7), the model will classify an email as spam only when it is very confident, reducing FP (fewer good emails lost) at the expense of slightly more FN (more spam gets through, but this is cheap). Letting spam through is far less harmful than blocking important emails.

Q5 Solution — Bias-Variance Tradeoff [3M]

(a) Diagnosis:

Model A (Train=500, Test=520): Both errors are high and similar $\Rightarrow$ **High Bias (Underfitting)**. The model is too simple to capture the underlying pattern.

Model B (Train=50, Test=900): Low train error but very high test error $\Rightarrow$ **High Variance (Overfitting)**. Huge gap (850) indicates memorization of training data.

(b) Fixes:

Model A fix: Increase model complexity — add more features, use polynomial features, switch to a more powerful model (e.g., from linear to ensemble). Decrease regularization if applied.

Model B fix: Add regularization (L1/L2), get more training data, use dropout (if neural network), reduce model complexity, apply cross-validation for hyperparameter tuning.

(c) Deployment Decision:

Deploy **Model A**. Although both perform poorly, Model A is **consistent** (train $\approx$ test), meaning its behavior is predictable and stable. Model B is unreliable — it performs well on training data but fails catastrophically on new data, making it dangerous in production. Model A's underfitting is easier to fix incrementally (add features/complexity) than Model B's overfitting, which may produce wildly incorrect predictions.

Q6 Solution — Decision Tree / Information Gain [3M]

(a) Parent Entropy:

10 fruits: 6 Apple, 4 Orange

$$\begin{aligned}
 H(S) &= -(6/10)\log_2(6/10) - (4/10)\log_2(4/10) \\
 &= -(0.6)\log_2(0.6) - (0.4)\log_2(0.4) \\
 &= -(0.6)(-0.737) - (0.4)(-1.322) \\
 &= 0.442 + 0.529 \\
 &= \mathbf{0.971 \text{ bits}}
 \end{aligned}$$

(b) Child Entropies:

Red (6 samples: 5A, 1O):

$$\begin{aligned}
 H(\text{Red}) &= -(5/6)\log_2(5/6) - (1/6)\log_2(1/6) \\
 &= -(0.833)(-0.263) - (0.167)(-2.585) \\
 &= 0.219 + 0.431 = \mathbf{0.650}
 \end{aligned}$$

Green (4 samples: 1A, 3O):

$$\begin{aligned}
 H(\text{Green}) &= -(1/4)\log_2(1/4) - (3/4)\log_2(3/4) \\
 &= -(0.25)(-2.0) - (0.75)(-0.415) \\
 &= 0.500 + 0.311 = \mathbf{0.811}
 \end{aligned}$$

Weighted Average:

$$\begin{aligned}
 H(S|\text{Color}) &= (6/10)(0.650) + (4/10)(0.811) \\
 &= 0.390 + 0.324 = \mathbf{0.714}
 \end{aligned}$$

(c) Information Gain:

$$IG(S, \text{Color}) = H(S) - H(S|\text{Color}) = 0.971 - 0.714 = \mathbf{0.257}$$

Yes, we should split on "Color." The IG of 0.257 is positive and substantial (reducing entropy by about 26%). Color is an informative feature for distinguishing apples from oranges. If this is the only feature, it becomes the root node. If there are other features, compare their IGs and choose the highest.